

Project Design Phase-I
Solution Architecture

Date	7 March 2026
Team ID	
Project Name	Visualization Tool for Electric Vehicle Charge and Range Analysis
Maximum Marks	4 Marks

Solution Architecture:

Solution architecture is a complex process that bridges the gap between business problems and technology solutions. Its goals in the context of Electric Vehicle (EV) analytics are to:

- Find the best technology solution to analyze EV charging patterns, battery performance, and driving range.
- Describe the structure, characteristics, behavior, and other aspects of the visualization tool to project stakeholders.
- Define features, development phases, and solution requirements for EV data analysis.
- Provide specifications according to which the solution is defined, managed, and delivered.

This architecture integrates data collection, secure transfer, storage, and visualization to empower planners, fleet managers, and consumers with actionable insights.

Example - Solution Architecture Diagram:

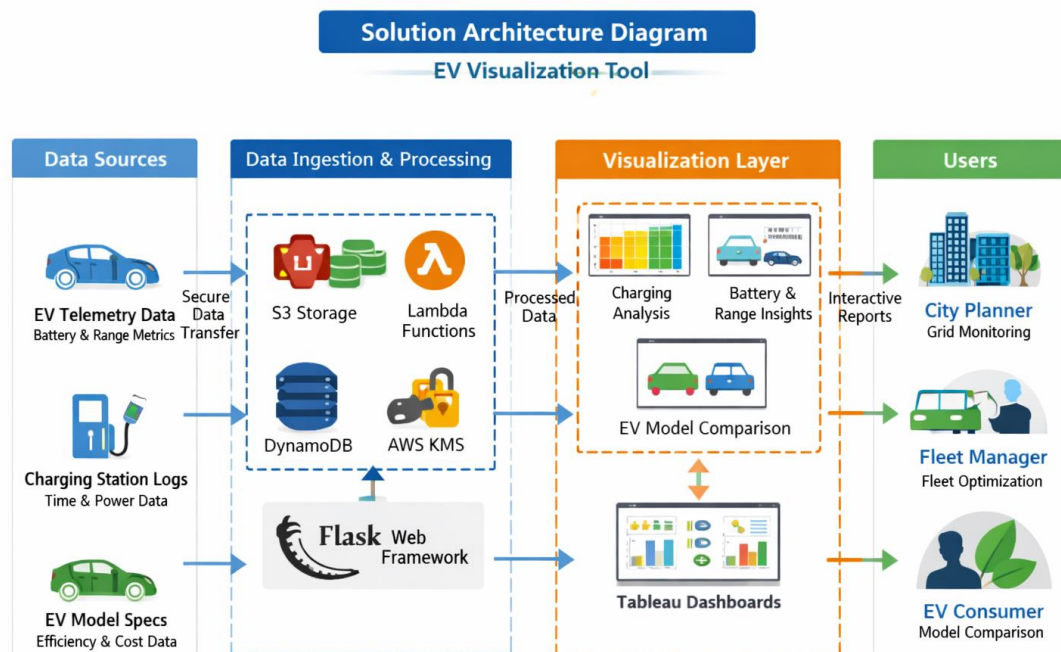


Figure 1: Architecture and data flow of the EV Charge & Range Analysis

Data Flow & Component Specifications:

- 1. EV Charging Stations and Sensors:** Collects data on how long the EV was charged, the battery's performance, and driving behaviour.
- 2. Automated Data Exports / API Integrations:** Provides relevant metric data and returns a structured file of data.
- 3. Raw Data (EV_ChargeData.csv):** Contains different variables related to the EV charging session such as charging time, the status of the battery, and an ID for the sessions.
- 4. Flask and ETL pipelines:** Processing data for visualisation by cleaning and transforming data.
- 5. Tableau Desktop (EVAnalytics.twb):** Creates dashboards and uses calculated fields like range efficiency.
- 6. Tableau Cloud/Server:** Provides an environment to host dashboards while protecting them.